

Categorical data analysis_GLM

R26121065吳思蓓

```
#Example Logistic Regression
diabete <- read.csv ( "https://gist.githubusercontent.com/chaityacshah/899a95deaf8b193
colnames(diabete)<-c("pregnant","glucose","blood_pressure","triceps",
                    "serum_insulin","mass","pedigree","age","diabetes")

## ##
fit.logit<- glm(diabetes~.,family=binomial(link=logit),data=diabete)
summary(fit.logit)
```

Call:

```
glm(formula = diabetes ~ ., family = binomial(link = logit),
    data = diabete)
```

Deviance Residuals:

Min	1Q	Median	3Q	Max
-2.5566	-0.7274	-0.4159	0.7267	2.9297

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-8.4046964	0.7166359	-11.728	< 2e-16 ***
pregnant	0.1231823	0.0320776	3.840	0.000123 ***
glucose	0.0351637	0.0037087	9.481	< 2e-16 ***
blood_pressure	-0.0132955	0.0052336	-2.540	0.011072 *
triceps	0.0006190	0.0068994	0.090	0.928515
serum_insulin	-0.0011917	0.0009012	-1.322	0.186065
mass	0.0897010	0.0150876	5.945	2.76e-09 ***
pedigree	0.9451797	0.2991475	3.160	0.001580 **

```

age                0.0148690  0.0093348   1.593 0.111192
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

(Dispersion parameter for binomial family taken to be 1)

```

Null deviance: 993.48  on 767  degrees of freedom
Residual deviance: 723.45  on 759  degrees of freedom
AIC: 741.45

```

Number of Fisher Scoring iterations: 5

```

#Example Logit Models for Nominal Responses
library(foreign)
data <- read.dta("https://stats.idre.ucla.edu/stat/data/hsbdemo.dta")
####
library(nnet)
data$prog2 <- relevel(data$prog, ref = "general")
fit.nomi <- multinom(prog2 ~ ses + write, data = data)

```

```

# weights:  15 (8 variable)
initial value 219.722458
iter  10 value 179.985215
final value 179.981726
converged

```

```
summary(fit.nomi)
```

```

Call:
multinom(formula = prog2 ~ ses + write, data = data)

```

```

Coefficients:
      (Intercept) sesmiddle  seshigh      write
academic -2.851973  0.5332914  1.1628257  0.05792480
vocation  2.366097  0.8246384  0.1802176 -0.05567514

```

```

Std. Errors:
      (Intercept) sesmiddle  seshigh      write
academic  1.166437  0.4437319  0.5142215  0.02141092

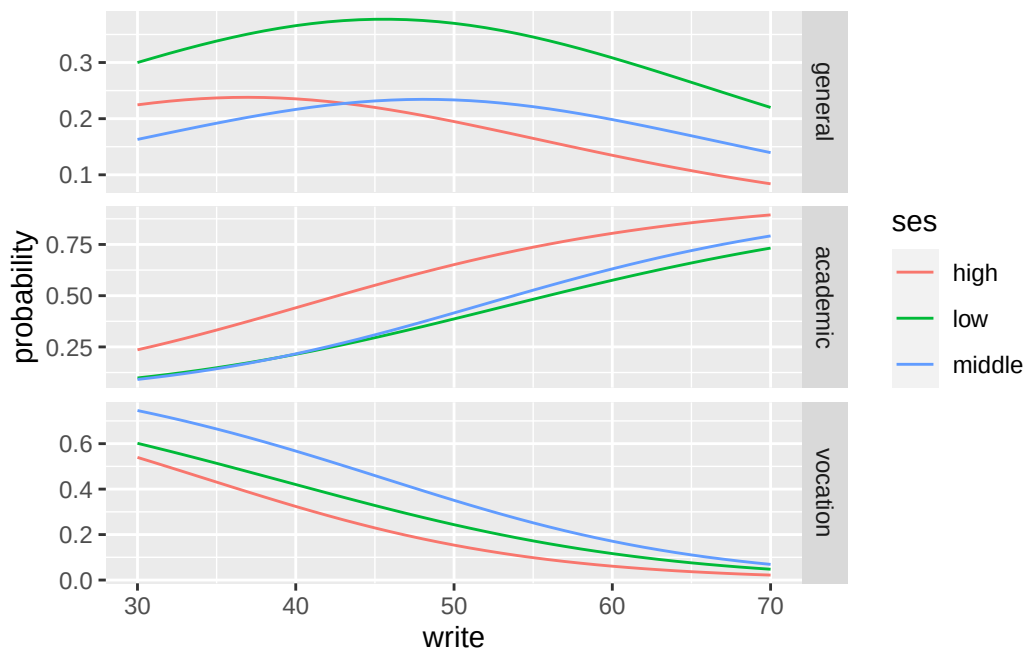
```

vocation 1.174251 0.4901237 0.6484508 0.02333135

Residual Deviance: 359.9635

AIC: 375.9635

```
## ##
library(reshape2)
library(ggplot2)
dwrite <- data.frame(ses = rep(c("low", "middle", "high"), each = 41),
                    write = rep(c(30:70), 3))
pp.write <- cbind(dwrite, predict(fit.nomi, newdata = dwrite,
                                type = "probs", se = TRUE))
lpp <- melt(pp.write, id.vars = c("ses", "write"),
            value.name = "probability")
ggplot(lpp, aes(x = write, y = probability, colour = ses)) + geom_line() +
  facet_grid(variable ~ ., scales = "free")
```



```
#Example Logit Models for Ordinal Responses
data <- read.dta("https://stats.idre.ucla.edu/stat/data/ologit.dta")
library(MASS)
fit.ordi <- polr(apply ~ pared + public + gpa, data = data, Hess=TRUE)
summary(fit.ordi)
```

Call:

```
polr(formula = apply ~ pared + public + gpa, data = data, Hess = TRUE)
```

Coefficients:

	Value	Std. Error	t value
pared	1.04769	0.2658	3.9418
public	-0.05879	0.2979	-0.1974
gpa	0.61594	0.2606	2.3632

Intercepts:

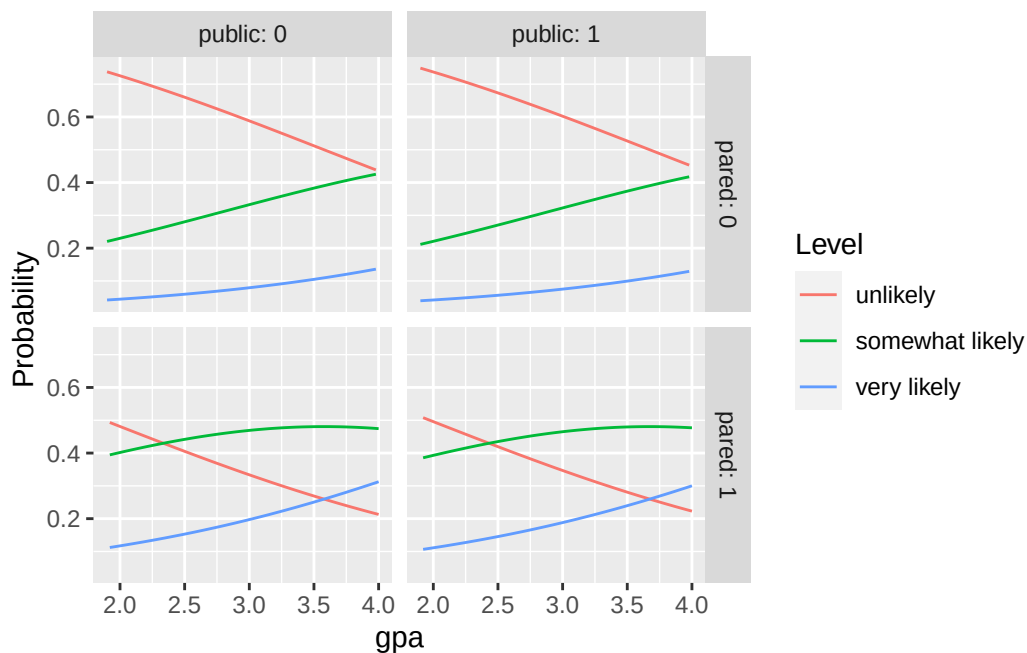
	Value	Std. Error	t value
unlikely somewhat likely	2.2039	0.7795	2.8272
somewhat likely very likely	4.2994	0.8043	5.3453

Residual Deviance: 717.0249

AIC: 727.0249

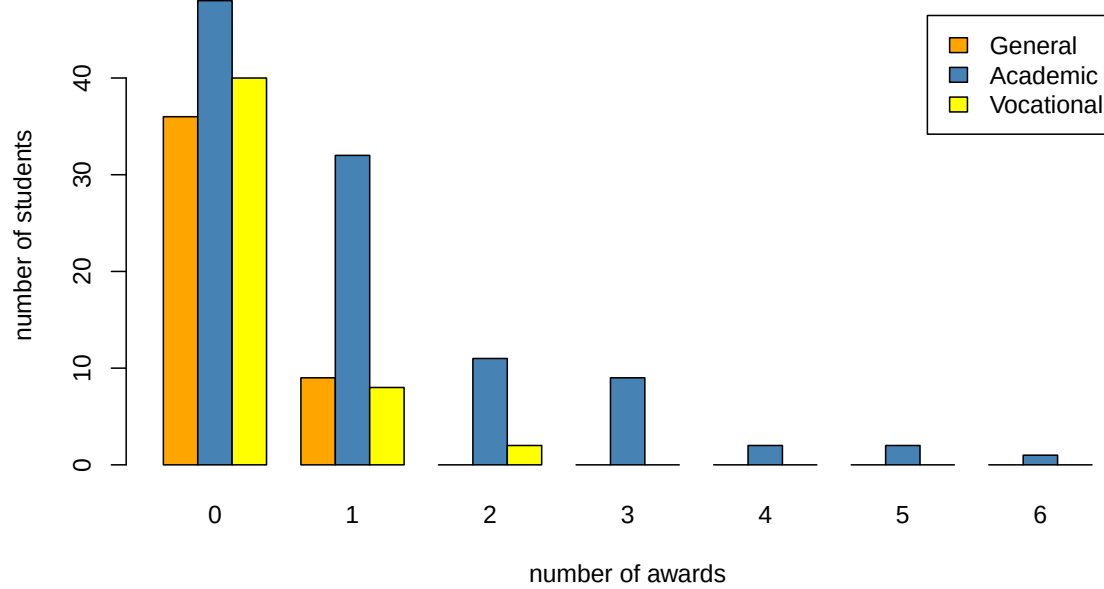
```
newdat <- data.frame(
  pared = rep(0:1, 200),
  public = rep(0:1, each = 200),
  gpa = rep(seq(from = 1.9, to = 4, length.out = 100), 4))

newdat <- cbind(newdat, predict(fit.ordi, newdat, type = "probs"))
lnewdat <- melt(newdat, id.vars = c("pared", "public", "gpa"),
  variable.name = "Level", value.name="Probability")
ggplot(lnewdat, aes(x = gpa, y = Probability, colour = Level)) +
  geom_line() + facet_grid(pared ~ public, labeller="label_both")
```

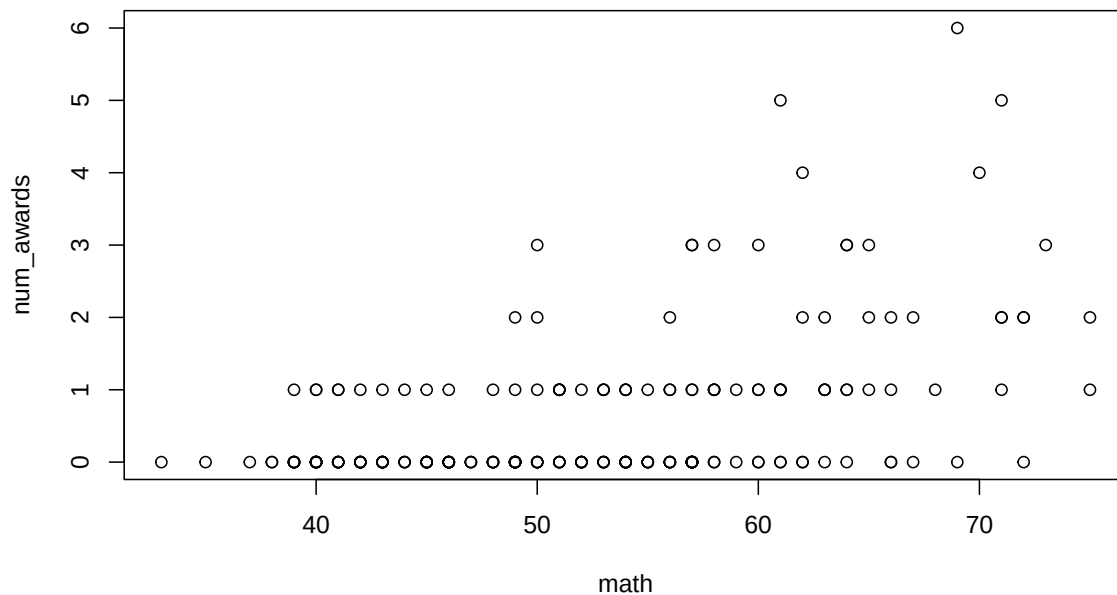


```
#Example Poisson Regression
#Y -> num_awards (      )
library(forcats)
awards <- read.csv ( "https://stats.idre.ucla.edu/stat/data/poisson_sim.csv" )
awards$prog<-fct_recode(as.character(awards$prog),'General'='1',
                        'Academic'='2','Vocational'='3')
awards$prog <- relevel(as.factor(awards$prog),ref="General")

## ##
barplot(height=t(table(awards$num_awards,awards$prog)),beside=T,
        col=c('orange','steelblue','yellow'),
        legend.text=c('General','Academic','Vocational'),
        xlab='number of awards',ylab='number of students')
```



```
plot(awards[,c(4,2)])
```



```
## ##
fit.poi <- glm(num_awards~math+prog,family=poisson(link='log'),data=awards)
summary(fit.poi)
```

Call:

```
glm(formula = num_awards ~ math + prog, family = poisson(link = "log"),
    data = awards)
```

Deviance Residuals:

Min	1Q	Median	3Q	Max
-2.2043	-0.8436	-0.5106	0.2558	2.6796

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-5.24712	0.65845	-7.969	1.60e-15 ***
math	0.07015	0.01060	6.619	3.63e-11 ***
progAcademic	1.08386	0.35825	3.025	0.00248 **
progVocational	0.36981	0.44107	0.838	0.40179

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

Null deviance: 287.67 on 199 degrees of freedom
Residual deviance: 189.45 on 196 degrees of freedom
AIC: 373.5

Number of Fisher Scoring iterations: 6

```
#Example Negative Binomial Regression
homicide <- data.frame(nvics=rep(c(0:6),2),race=rep(c('Black','White'),each=7),
                      Freq=c(119,16,12,7,3,2,0,1070,60,14,4,0,0,1))
## mean, variance
black <- c()
white <- c()
for(i in 0:6){
  black <- c(black,rep(i,homicide$Freq[i+1]))
  white <- c(white,rep(i,homicide$Freq[i+8]))
}
mean(black)
```

```
[1] 0.5220126
```

```
var(black)
```

```
[1] 1.149829
```

```
mean(white)
```

```
[1] 0.09225413
```

```
var(white)
```

```
[1] 0.1552448
```

```
##if use poisson regression##
fit.poi <- glm(nvics~race,family=poisson(link='log'),weight=Freq,data=homicide)
summary(fit.poi)
```

Call:

```
glm(formula = nvics ~ race, family = poisson(link = "log"), data = homicide,
     weights = Freq)
```

Deviance Residuals:

Min	1Q	Median	3Q	Max
-14.051	0.000	5.257	6.216	13.306

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.6501	0.1098	-5.922	3.17e-09 ***
raceWhite	-1.7331	0.1466	-11.825	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

Null deviance: 962.80 on 10 degrees of freedom
 Residual deviance: 844.71 on 9 degrees of freedom
 AIC: 1122

Number of Fisher Scoring iterations: 6

```
##Use NegBin Regression##
library(MASS)
fit.nb <- glm.nb(nvics~race,weight=Freq,data=homicide)
summary(fit.nb)
```

Call:

```
glm.nb(formula = nvics ~ race, data = homicide, weights = Freq,
        init.theta = 0.2023119205, link = log)
```

Deviance Residuals:

Min	1Q	Median	3Q	Max
-12.754	0.000	2.086	3.283	9.114

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-0.6501	0.2077	-3.130	0.00175	**
raceWhite	-1.7331	0.2385	-7.268	3.66e-13	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for Negative Binomial(0.2023) family taken to be 1)

Null deviance: 471.57 on 10 degrees of freedom
 Residual deviance: 412.60 on 9 degrees of freedom
 AIC: 1001.8

Number of Fisher Scoring iterations: 1

Theta:	0.2023
Std. Err.:	0.0409

2 x log-likelihood: -995.7980

```
#Example Zero-inflated Model
fish <- read.csv("https://stats.idre.ucla.edu/stat/data/fish.csv")
fish$count2<-fish$count
fish$count2[fish$count2>5]<-">5"
table(fish$count2)[c(2:7,1)]
```

0	1	2	3	4	5	>5
142	31	20	12	6	10	29

```
##zero-inflated model##
library(pscl)
fit.zero <- zeroinfl(count~child+camper+persons,data=fish)
summary(fit.zero)
```

Call:

```
zeroinfl(formula = count ~ child + camper + persons, data = fish)
```

Pearson residuals:

	Min	1Q	Median	3Q	Max
	-3.05440	-0.74336	-0.44275	-0.07559	27.99304

Count model coefficients (poisson with log link):

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.79826	0.17081	-4.673	2.96e-06 ***
child	-1.13666	0.09299	-12.224	< 2e-16 ***
camper	0.72425	0.09314	7.776	7.51e-15 ***
persons	0.82904	0.04395	18.862	< 2e-16 ***

Zero-inflation model coefficients (binomial with logit link):

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	1.6636	0.5155	3.227	0.00125 **
child	1.9046	0.3261	5.840	5.21e-09 ***
camper	-0.8336	0.3527	-2.364	0.01808 *
persons	-0.9228	0.1992	-4.632	3.62e-06 ***

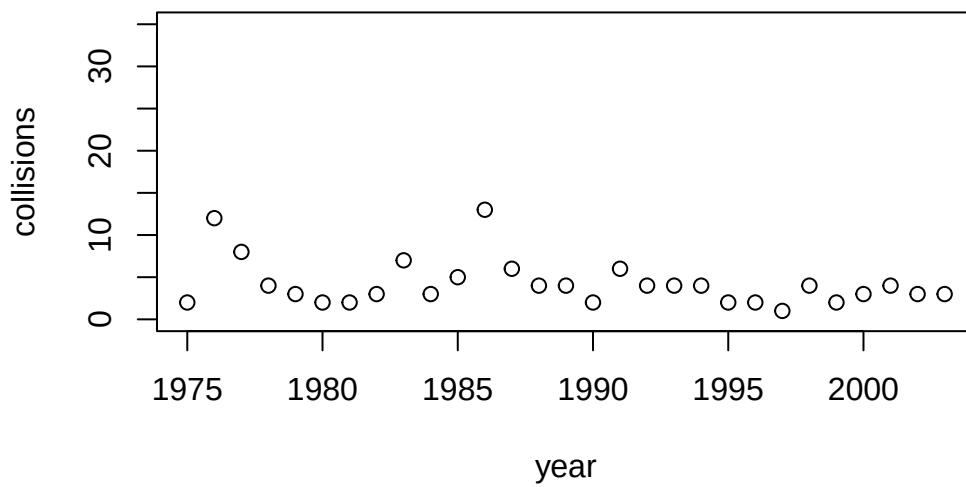
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Number of iterations in BFGS optimization: 12

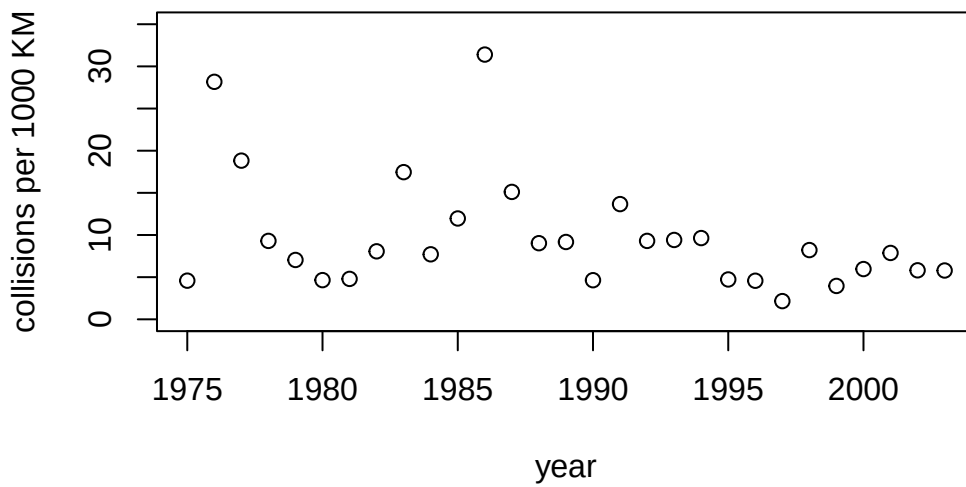
Log-likelihood: -752.7 on 8 Df

```
#Example models for rate
year <- seq(2003,1975,by=-1)
KM <- c(518,516,508,503,505,487,463,437,423,415,425,430,439,
        431,436,443,397,414,418,389,401,372,417,430,426,430,
        425,426,436)
TrRd <-c(3,3,4,3,2,4,1,2,2,4,4,4,6,2,4,4,6,13,5,3,7,3,2,2,3,
        4,8,12,2)
train <- data.frame('Year'=year,'KM'=KM,'TrRd'=TrRd)

## ##
plot(year,TrRd,ylab='collisions',ylim=c(0,35)) #
```



```
plot(year,1000*TrRd/KM,ylab='collisions per 1000 KM',ylim=c(0,35)) #
```



```
##fit poisson model##
fit.rate <- glm(TrRd~I(Year-1975),offset=log(KM),family=poisson(link='log'),
               data=train)
summary(fit.rate)
```

Call:

```
glm(formula = TrRd ~ I(Year - 1975), family = poisson(link = "log"),
    data = train, offset = log(KM))
```

Deviance Residuals:

Min	1Q	Median	3Q	Max
-2.0580	-0.7825	-0.0826	0.3775	3.3873

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-4.21142	0.15892	-26.50	< 2e-16 ***
I(Year - 1975)	-0.03292	0.01076	-3.06	0.00222 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

Null deviance: 47.376 on 28 degrees of freedom
 Residual deviance: 37.853 on 27 degrees of freedom
 AIC: 133.52

Number of Fisher Scoring iterations: 5